

FITBUDDY – AI FITNESS PLAN GENERATOR USING GEMINI MODELS

7. Project Documentation Phase

The Project Documentation Phase records the complete development process of FitBuddy so that the project can be understood, maintained, evaluated, and demonstrated in the future.

The documentation begins with the project title and introduction. It explains the purpose of FitBuddy and the problem that the application attempts to address. The objectives of the application are also described.

The requirement analysis section documents the user inputs, processing requirements, outputs, functional requirements, and technologies used.

The system design section explains the architecture of FitBuddy. It describes how the frontend communicates with the FastAPI backend, how the backend communicates with the Gemini AI modules, and how user information and generated plans are stored in SQLite.

The implementation section describes the major modules of the project. These include the FastAPI application, database configuration, data models, Pydantic schemas, Gemini workout generator, nutrition generator, feedback/update module, HTML templates, and CSS.

The documentation also explains the workflow of the application. The user first enters their fitness information. FitBuddy validates the information and generates a personalized workout plan and nutrition/recovery recommendation. The user can then provide feedback, after which the application generates an updated plan.

Screenshots of important stages can be included in the documentation, such as the home page, user input form, generated workout plan, nutrition recommendation, feedback form, updated workout plan, and user/admin view.

The testing section records the tests performed and the corresponding results. Installation instructions, environment configuration, Gemini API-key configuration, and instructions for running the FastAPI application can also be included.

The final documentation therefore acts as a complete reference for understanding, installing, running, testing, and demonstrating the FitBuddy project.